AP PRECALCULUS

Unit 1 • Lesson 1.8 practice • Answer key

# Lesson 1.8 • Rational Functions and Zeros

5 multiple-choice questions and 1 free-response question • Answer key with explanations

**Directions**

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

## Question 1

NO CALCULATOR

What are the zeros of $r(x) = \dfrac{x^2 - 9}{x + 2}$?

(A) $-2$

(B) 3 and $-3$

(C) 9

(D) 3 only

**Solution**

**Answer: (B).** The numerator is 0 at $x = \pm 3$, and neither value makes the denominator 0.

## Question 2

NO CALCULATOR

What are the zeros of $r(x) = \dfrac{x^2 - 4x}{x - 4}$?

(A) 0 and 4

(B) 4 only

(C) 0 only

(D) There are none.

**Solution**

**Answer: (C).** $x = 4$ is not in the domain of $r$, so only $x = 0$ is a zero.

## Question 3

NO CALCULATOR

What are the zeros of $r(x) = \dfrac{(x + 1)(x - 5)}{(x - 5)(x + 3)}$?

(A) $-1$ only

(B) $-1$ and 5

(C) 5 only

(D) $-3$ and $-1$

**Solution**

**Answer: (A).** $x = 5$ is excluded from the domain (there is a hole), and $x = -3$ is a zero of the denominator.

### Question 4

NO CALCULATOR

What is the solution set of $\dfrac{x+2}{x-1} < 0$?

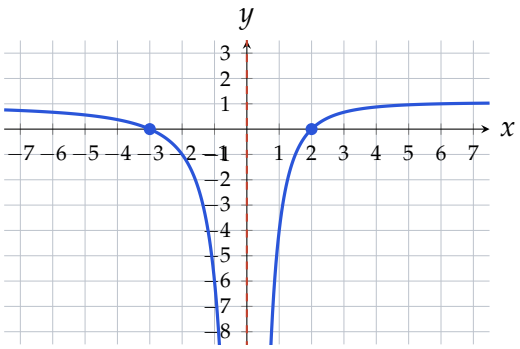
(A) $(-\infty, -2)$
(B) $(1, \infty)$
(C) $(-\infty, -2) \cup (1, \infty)$
(D) $(-2, 1)$

**Solution**
**Answer: (D).** The sign changes at $-2$ and $1$. Test $x = 0$: $\frac{2}{-1} < 0$. Outside the interval the quotient is positive.

### Question 5 (graph)

NO CALCULATOR

The graph of $r(x) = \dfrac{(x+3)(x-2)}{x^2}$ is shown. It has zeros at $x = -3$ and $x = 2$ and a vertical asymptote $x = 0$. What is the solution set of $r(x) > 0$?




(A) $(-3, 2)$
(B) $(-3, 0) \cup (0, 2)$
(C) $(-\infty, -3) \cup (2, \infty)$
(D) $(2, \infty)$

**Solution**
**Answer: (C).** The graph is above the $x$-axis to the left of $x = -3$ and to the right of $x = 2$. Between them it is below the axis on both sides of the asymptote.

### Question 6 (Free response)

NO CALCULATOR

Let $r(x) = \dfrac{x^2 - x - 2}{x^2 - 9}$.

(a) Factor the numerator and denominator, and find the zeros of $r$.
(b) Make a sign chart for $r$ and solve $r(x) > 0$.

**Solution**
(a) $r(x) = \dfrac{(x-2)(x+1)}{(x-3)(x+3)}$. No factor cancels. Zeros: $x = 2$ and $x = -1$.
(b) The sign can change at $-3, -1, 2, 3$. Test values: $x = -4$: $\frac{(-6)(-3)}{(-7)(-1)} > 0$; $x = -2$: $\frac{(-4)(-1)}{(-5)(1)} < 0$; $x = 0$: $\frac{(-2)(1)}{(-3)(3)} > 0$; $x = 2.5$: $\frac{(0.5)(3.5)}{(-0.5)(5.5)} < 0$; $x = 4$: $\frac{(2)(5)}{(1)(7)} > 0$.
So $r(x) > 0$ on $(-\infty, -3) \cup (-1, 2) \cup (3, \infty)$.